

Consider the linear optimization problem

$$\min_{x \in \mathbb{R}^2} x_1$$

subject to

$$x_1, x_2 \geq 0.$$

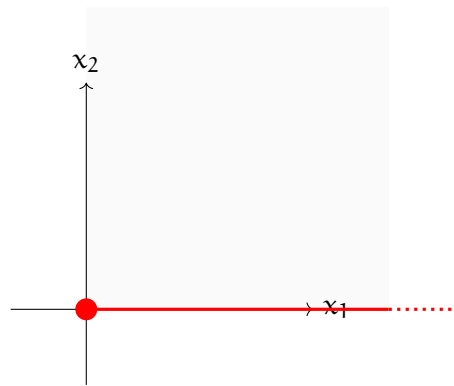
For any $\alpha \geq 0$, the solution

$$\begin{pmatrix} 0 \\ \alpha \end{pmatrix}$$

is optimal. Indeed, it is feasible, and x_1 achieves its smallest possible value. Therefore, the problem has an infinite number of optimal solutions. The feasible set is the non negative quadrant. It has only one vertex:

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

as illustrated in the following figure:



Consequently, the claim is not true.